Preeti Verma

Girona, Spain

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EDUCATION

Eötvös Loránd University (ELTE); University of Girona (UdG) **Sept. 2022 - June 2024**
Erasmus Mundus Master in Intelligent Field Robotic Systems (IFRoS) *Hungary, Spain*

ZHCET, Aligarh Muslim University (AMU) **Aug. 2017 – Jul. 2021**
Bachelor of Technology in Electrical Engineering; GPA: 9.24/10 (in top 5%) *Aligarh, India*

TECHNICAL SKILLS

Programming Languages: Python (Proficient), C++ (Intermediate)
Software Skills: ROS, Gazebo, MATLAB/Simulink, Microsoft Visio, Proteus, LaTeX, Microsoft Office
Libraries and Frameworks:
NumPy, pandas, Shapely, scikit-learn, open3d, OMPL, py_trees, SciPy, OpenCV, PIL, simpleITK
Deep Learning: Proficient in PyTorch, TensorFlow
Reinforcement Learning: Stable Baselines, OpenAI Gym
Tools: Git, Docker; **Protocols:** CAN bus; **Operating Systems:** Linux, Windows
Sensors: 2D/3D LiDAR, Event-based/Standard Cameras, GPS, IMU.

WORK EXPERIENCE

Artificial Intelligence Researcher **January 2025 – Present**
MICELab, UdG *Girona, Spain*

- Contributing to the EU-funded Prometeus project on safe, personalized glucose control for neonates.
- Designed adaptive MPC/PID controllers with patient-specific models, uncertainty-aware tuning, and real-time deployment in clinical settings.
- Fused 10 clinical data streams with AI-extracted nutritional signatures from Gemini-parsed logs. Applied Multi-View Canonical Correlation Analysis (mCCA) to identify high-resolution metabolic phenotypes for precision GLP-1 therapy.

Research Assistant **November 2021 – June 2022**
Non-Conventional Energy Laboratory, AMU *Aligarh, India*

- Conducted seminars on Typhoon HIL training for undergraduate students.
- Performed research on optimal maximum power point tracking for photovoltaic systems using Typhoon HIL, focusing on optimization through metaheuristic algorithms.
- Co-supervised Master's thesis and guided Bachelor's students in research projects.

RESEARCH EXPERIENCE

Autonomous Navigation using Deep Reinforcement Learning (DRL) **Feb 2024 – June 2024**
Master thesis student at UdG *Girona, Spain*

- Developed a custom Gymnasium environment tailored to test DRL algorithms in autonomous navigation scenarios.
- Implemented a LiDAR data downsampling method to reduce computational costs while maintaining navigation efficiency.
- Conducted a comparative analysis of Soft Actor-Critic and Proximal Policy Optimization algorithms, using the Stable-Baselines3 library, providing insights into their performance in various environments.
- Full thesis

Medical Imaging Quality Assessment with Radiomics **July 2023 – August 2023**
Research intern at BCN-AIM lab, University of Barcelona *Barcelona, Spain*

- Evaluated synthetic data generation techniques, emphasizing clinically relevant features and biomarkers, addressing the lack of standardized evaluation metrics.
- Investigated Radiomics Fréchet inception distance (FID) and FID ratio robustness against normalization, comparing them with regular FID metrics.
- Explored the influence of image quality on Radiomics FID performance, particularly with lower-quality images. Outcome: Established the proposed metric's heightened sensitivity to noise, indicating its potential in noisy imaging environments.
- Examined the correlation between Radiomics FID and downstream task performance, assessing its predictive value in clinical decision-making.

- Utilized various libraries, including os, numpy, torch, PIL, scipy, SimpleITK, Pyradiomics, and other important libraries for comprehensive data analysis.

Autonomous Robotic Explorer and Manipulator — February 2023 – July 2023

Master student at UdG — *Girona, Spain*

- Engaged in a 2-month project collaboratively, focusing on transforming a differential drive robot equipped with a 4-DOF robotic arm into an autonomous explorer and mobile manipulator using ROS (Robot Operating System).
- Took an active role in the project by contributing to the implementation of a frontier-based exploration algorithm, merging Rapidly Exploring Random Trees and the Dynamic Window Approach. This solution was rigorously tested in both Gazebo and Stonefish simulator environments. [Planning project]
- Played an integral part in developing Pose-based EKF SLAM, integrating Iterative Closest Point (ICP) scan matching for real-time localization and mapping using data from LiDAR odometry and IMU sensors. [Localization project]
- Implemented an efficient event-based feature tracking methodology using the ICP algorithm for Dynamic and Active-pixel Vision sensor (DAVIS) data, optimizing computational processing for high-frequency event streams. [Perception project]
- Participated in enhancing the robot's perception capabilities by integrating a RealSense camera and implementing an ArUco marker detection pipeline to achieve precise object identification.
- Collaborated in designing and developing Task Priority Redundancy Resolution algorithms, ensuring efficient and secure manipulation of objects, including tasks like object picking, transportation, and placement. [Intervention project]
- Contributed to the collaborative effort to achieve autonomous decision-making, leveraging the power of Behavior Trees through the py_trees library. This effort resulted in a robot capable of making intelligent and seamless decisions in its operations. [Project Video1, Project Video2]

Multi-Span Medical Question Answering — June 2022 – August 2022

Research Intern at IIT Patna — *Aligarh, India*

- Team of five developed a transformer-based query semantic and knowledge-guided multi-span question-answering model (QueSemKnow) for Medical Question Answering (MedQA) tasks.
- Proposed a two-phased approach: a multi-task model for extracting query semantics, including intent identification and question type prediction, and context selection and answer extraction based on semantic information and relevant knowledge from a knowledge graph.
- Built a semantically aware medical question-answering corpus called QueSeMSpan MedQA, tagging each question with its semantic information.
- Demonstrated the effectiveness of the QueSemKnow model by outperforming several baselines and existing state-of-the-art models on multiple datasets.

A Computer Vision-based Disease and Quality Prediction Technique — June 2021 – July 2021

Research group member — *Aligarh, India*

- Team of five researched the classification of potatoes and lemons based on their quality into three classes, i.e., healthy, diseased, and rotten.
- Utilized a quality-based classification technique using a Convolutional Neural Network, followed by several computer vision procedures like colour, shape, and texture extraction.
- Responsible for data curation and software visualization.
- Achieved accuracy of 94.1% for classification.

PROFESSIONAL CERTIFICATIONS

Improving Deep Neural Networks: Hyperparameter tuning, Regularization & Optimization — Dec 2025

Coursera (Deep Learning Specialization, Course 2) — *Online*

- Mastered advanced techniques for improving deep learning model performance, focusing on optimization algorithms (Momentum, Adam, RMSprop), regularization (L2, dropout), and robust hyperparameter tuning strategies.

Neural Networks and Deep Learning — Nov 2025

Coursera (Deep Learning Specialization, Course 1) — *Online*

- Gained a strong foundational understanding of neural networks and deep learning, covering computation graphs, forward/backward propagation, and the implementation of fully connected deep neural networks.

Certification Course On Machine Learning — July 2020

Indian Institute of Technology (IIT) Kanpur, E & ICT Academy — *Online*

- Comprehensive training in foundational machine learning algorithms, covering theoretical concepts and practical applications of predictive modeling.

PUBLICATIONS

- N. Konz, R. Osuala, **P. Verma**, Y. Chen, H. Gu, H. Dong, Y. Chen, A. Marshall, L. Garrucho, K. Kushibar, D. M. Lang, G. S. Kim, L. J. Grimm, J. M. Lewin, J. S. Duncan, J. A. Schnabel, O. Diaz, K. Lekadir, and M. A. Mazurowski, "Fréchet Radiomic Distance (FRD): A Versatile Metric for Comparing Medical Imaging Datasets," *Medical Image Analysis*, accepted, 2025.
- R. Osuala, D. Lang, **P. Verma**, S. Joshi, A. Tsirikoglou, G. Skorupko, K. Kushibar, L. G. Morras, W. L. Pinaya, O. Diaz, J. A. Schnabel, and K. Lekadir, "Towards Learning Contrast Kinetics with Multi-Condition Latent Diffusion Models," 27th International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI 2024), Oct. 6 - Oct. 10, 2024, Marrakesh, Morocco.
- A. Tiwari, S. Bera, **P. Verma**, J. V. Manthena, S. Saha, P. Bhattacharyya, M. Dhar, and S. Tiwari, "Seeing is believing! Towards Knowledge-Infused Multi-modal Medical Dialogue Generation," 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024), May 20 - May 25, 2024, Torino, Italy.
- **P. Verma**, A. Alam, A. Sarwar, M. Tariq, H. Vahedi, D. Gupta, S. Ahmad, and A. Shah Noor Mohamed, "Meta-Heuristic Optimization Techniques Used for Maximum Power Point Tracking in Solar PV System," *Electronics*, vol. 10, no. 19, p. 2419, Oct. 2021.

PROJECTS

Pomona 3D Graph SLAM Online Environment Mapping **September 2023 – December 2023**
Robotics Project *Budapest, Hungary*

- Led a 2-member team in implementing the Pomona 3D Graph SLAM online environment mapping project, employing ROS as the primary implementation platform.
- Integrated sensor data from 3D LiDAR, GPS, IMU, and CAN bus with ROS, utilizing advanced sensor fusion techniques to enhance pose graph accuracy and optimize mapping results for the Pomona robot.
- Customized and fine-tuned algorithms within the hdl_graph_slam GitHub repository, tailoring them specifically for the ROS framework and the unique requirements of the Pomona robot.
- Conducted comprehensive comparative analyses of mapping performance using GPS only, IMU only, and their combination within the ROS ecosystem, emphasizing the importance of ROS in facilitating seamless integration and communication among system components.

Meta-heuristic Optimization Algorithms **August 2020 – May 2021**
Undergraduate Final Year Project *Aligarh, India*

- Led a 3-member team working on meta-heuristic optimization algorithms for tracking the maximum power point of PV arrays under partial shading conditions.
- Reviewed 18 optimization techniques and executed simulations in MATLAB/Simulink for Particle Swarm Optimization, Cuckoo Search, and the Jaya algorithm.
- Critiqued the algorithms based on various parameters such as complexity, tracking speed, tracking accuracy, steady-state oscillations, etc., and documented that the Jaya algorithm outperformed the other two in terms of tracking accuracy.

CONFERENCES & WORKSHOPS

EuroPython 2025 **July 2025**
Conference Attendee, EuroPython Society *Prague, Czech Republic*

- Attended talks, tutorials, and community events to stay up to date with the latest developments in the Python ecosystem, AI/ML tooling, and best practices.

ELLIS Winter School on Foundation Models (FoMo 2025) **March 2025**
Participant, ELLIS Unit Amsterdam *Amsterdam, Netherlands*

- Attended advanced lectures on scaling laws, lifelong learning, multimodal systems, and generative AI from researchers at LAION, DeepMind, EPFL, and other leading institutions, and explored recent research in poster sessions led mainly by PhD and postdoctoral researchers.

Unmanned Aerial Vehicles and Drones **March 2019**
University Polytechnic (Boys), AMU *Aligarh, India*

- Participated in a three-day workshop on unmanned aerial vehicles and drones.
- Gained insight into drone design and development through practical sessions.

Radio Controlled Aircraft Design **March 2019**
Delhi Technological University *Delhi, India*

- Attended a two-day project-based training program on radio-controlled aircraft design
- Devised the electric RC plane with four group members and contributed to mathematical computations and designing for plane construction from foam.

LANGUAGE PROFICIENCY

- English (Fluent)
- Spanish (Beginner)
- Hindi (Native)

POSITION OF RESPONSIBILITY

- **Reviewed** two articles related to optimization algorithms for Scientific Reports, an open-access journal, in 2022.
- Served as a **Research Assistant** at AMU for three months on a project on real-time optimization algorithm implementation using Typhoon HIL in collaboration with UNSW Canberra, Australia.
- Supervised a five-member team for organizing Vazeto and Picturesque events in College Fest ZARF 2019.
- Elected as the House-captain of my school in 2014 and 2015, consecutively. My role was to communicate the relevant news to house members, motivate the students to participate in extracurricular activities, and assist with solidarity campaigns and events.

ACHIEVEMENTS

- Awarded the Global Korea Scholarship (GKS), the Banach Scholarship for a Master's in Computer Science, and the Erasmus Mundus Joint Master's Scholarship for the IFRoS program, based on academic merit and demonstrated potential in the field.
- Participated in EMBARK 2021 organized by Upraised and qualified to the pre-final round among top 2% performers out of 4,55,534 participants in design track.
- Awarded AWS Machine Learning Scholarship in 2021. This scholarship program aimed to upskill participants in machine learning and cultivate the next generation of ML practitioners.
- Awarded Chiragdeep Scholarship in 2017. This scholarship promotes girls to pursue higher education goals by providing monthly stipends that abide by the academic excellence for full degree programs.
- Took part in Women's Power Race of 2.5 km on Home-guard Day Celebration in 2014 and obtained the tenth rank among 250 candidates.
- Granted Merit Certificate for securing above the 50th percentile in the National Financial Literacy Assessment Test, 2014 in India.

REFERENCES

Dr. Narcis Palomeras Rovira
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